Aakash **Gurung**

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Education

Expected Dec 2026 — M.A. and B.S. in Mathematics; Minor: Digital, Public, and Professional Writing, The University of Alabama, Tuscaloosa, AL

∞ Research Interests

I am broadly interested in **Algebraic Combinatorics**, **Applied Mathematics**, and **Geometry**.

∑ Research Experience

Present – Feb 2026 — **Rectifiability, University of Alabama,**
Mentor: Professor Simon Bortz
- Along with a postdoctoral researcher, studying rectifiability characterization in the parabolic setting and connections to Riesz transforms and tangent measures.

Geometric Measure Theory | Rectifiability

Present – Feb 2025 — **Combinatorial Perspective of (q,t)-Catalan Symmetry, University of Alabama,**
Mentor: Professor Kyungyong Lee
- Working on a subset of the combinatorial proof of the (q,t)-Catalan symmetry.

Algebraic Combinatorics | Catalan Numbers

October 2026 – March 2026 — **LLM Distillation, University of Alabama,**
Mentors: Professor Chuntian Wang and Professor Jia Zhao
- Along with a graduate student, worked on LLM distillation in network-based epidemic simulation.

LLM Distillation | Epidemic Modeling | Network-Based Simulation

Aug 2026 – June 2026 — **Multi-Agent AI for Mathematical Reasoning, RIPS, UCLA,**
Academic Mentor: Mehmet Yigit Turali
Industry Sponsor: OpenAI
Industry Mentors: Giambattista Parascandolo (GB) and Professor Gregory Valiant
- Along with three fellow research interns, developed and evaluated multi-agent AI systems for mathematical reasoning and proof-oriented research workflows. Applied the system to a research problem under Professor Kyungyong Lee's guidance.

AI for Mathematics | Mathematical Reasoning | Multi-Agent Systems

Dec 2025 – May 2025 — **Finite-Size Effects in Epidemic Models, University of Alabama Mathematics Summer REU,**
Advisors: Professors Chuntian Wang, Yuanyuan Song, Yuanzhen Shao
- Along with four fellow student researchers, worked on developing and analyzing the dynamics of an epidemic compartment model (SIHRS), incorporating vital dynamics through independent Poisson clocks and a martingale formulation. Used real-world county data from COVID-19 to show the model's usefulness. Also worked on the global stability analysis of the model.

Applied Mathematics | Epidemic Modeling | Stochastic Analysis

June 2024 – Feb 2024 — **Game of Cycles on Maximal Plane Graphs, CUNY Research Scholars Program,**
Advisor: Professor Malgorzata Marciniak
- Defined "IO Maximal Plane Graphs" and analyzed invariant properties to determine the game outcome.
- Established that the winning strategy is determined by the parity of the graph's vertices.

Combinatorial Game Theory | Graph Theory

Aug 2023 – June 2023 — **Method of Brackets and Bessel Function Integrals, Polymath Jr 2023,**
Advisor: Professor Victor H. Moll
- Applied the "Method of Brackets" to provide rigorous proofs for entries involving Bessel functions of the first and second kind from the Gradshteyn and Ryzhik tables.

Special Functions | Integral Calculus | Bessel Functions

Aug 2023 – Mar 2023 — **Continued Fractions, a-Fibonacci Numbers, and Middle b-Noise, Independent Project,**
Advisor: Professor Cheng Han Pan
- Generalized palindromic continued fractions $[1,\ldots,1,3,1,\ldots,1]$ to $[a,\ldots,a,b,a,\ldots,a]$ using a-Fibonacci sequences.
- Showed that the a-th metallic ratio limit is invariant under the middle noise term b.

Number Theory | Continued Fractions | Fibonacci Sequences

Publications

To appear, 2026 **A. Gurung**, S. Wagle, A. Carr, C. McCann, K. Kodatt, Y. Song, Y. Shao, C. Wang. "An exploration of finite-size effects in the dynamics of epidemic compartmental modelling." *European Journal of Applied Mathematics*.

Expository/Other Works

2024 **A. Gurung** and C.-H. Pan. "Continued Fractions, a-Fibonacci numbers, and the middle b-noise," *Mathematics Exchange*, 18(1), 77–87.

2024 (with the Polymath Jr. Group). "The integrals in Gradshteyn and Ryzhik. Part 34: Bessel functions," *Scientia Series A: Mathematical Sciences*, 34, 109–129.

Conferences & Workshops

November 2026 Algecom-XXVIII, **Poster Presentation**
Supported by NSF ALGECOM and UIUC RTG

October 2026 SIAM CSS 2026, Jackson State University, **Invited Talk**
Partially Supported by ASSURE

March 2026 AMS Southeastern Sectional Meeting, **Presenter**
Supported by ASSURE

May 2024 CUNY Undergraduate Research Day 2024, **Presenter**

Aug 2024 MathFest 2024, **Presenter**

Sept–Nov 2024 Preliminary Arizona Winter School 2024: Symmetries of Root Systems, Attendee

Honors & Awards

2026 Thomas W. Palmer Math Award, University of Alabama

2026 Randall Outstanding Undergraduate Research Award, University of Alabama

2025 ASSURE Grant, University of Alabama

2024 Best Poster Award, CUNY Undergraduate Research Day

2023 Samuel J. Steinberger, Jr. Memorial Award, Juniata College

2020–2021 Nepal Mathematical Olympiad (Top 12); Nepal Astronomy Olympiad (Rank 1); USA Astronomy & Astrophysics Competition (National Qualifier)

Work Experience

May 2026 – Aug 2025 | **IT Service Desk Student Assistant, University of Alabama, Tuscaloosa, AL**
- Provide timely software and technology support to resolve user issues efficiently.

IT Support | Technical Support

May 2025 – Sept 2024 | **Peer Tutor, Mathematics Technology Learning Center, Tuscaloosa, AL**
- Drop-in tutor for Calculus 1, 2, and 3.
- Run recitation classes for Calculus 1.

Calculus | Mathematics Education | Tutoring

Skills

Programming Python (NumPy, Pandas, SciPy), Julia, MATLAB, JavaScript, HTML, CSS

Tools PowerQuery, LaTeX, Git

Other Grant Writing